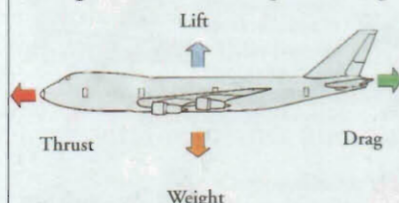


Forces of flight

An aircraft needs two forces to fly: lift to keep it up and thrust to propel it forward. Lift overcomes the plane's weight, and thrust overcomes the drag caused by the air flowing past the plane. When an aircraft is cruising, lift is equal to weight and thrust is equal to drag.

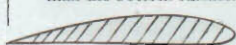


Wings

An aircraft's wings create lift. To do this, they need air to flow over them.

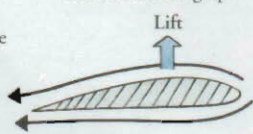
The aerofoil shape

If you cut an aircraft wing in two and looked at the end, you would see a special cross-section called an aerofoil. The top surface is longer and more curved than the bottom surface.



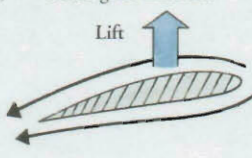
The aerofoil at work

The air pressure beneath the wing is greater than above it, and lifts the wing up.



Angle of attack

Tilting the angle of the blades gives extra lift.



Flying controls

An aircraft is steered through the air by way of three main control surfaces – the elevators on the tailplane, the ailerons on the wings, and a rudder on the fin.



Elevators make the aircraft's nose tilt up and down.



Ailerons make the aircraft roll from side to side.



The rudder makes the aircraft "yaw" to left or right.

Aero engines

An aircraft's engines drive it through the air by producing thrust. Different types of engine produce thrust in different ways. Piston and turbo-prop engines drive propellers that screw into the air, just as a ship propeller bites into water. Turbo-jet and turbo-fan engines produce a fast-moving stream of gas which pushes the aircraft forwards.

Piston engines

These work in the same way as car engines. Petrol and air vapour are mixed in the engine's cylinders and they cause an explosion. The explosions push pistons, which turn a shaft. The shaft then turns a propeller.



Turbo-prop engines

The simplest type of jets – a turbo-jet engine with a propeller is called a turbo-prop engine. A motor turns the compressor and the propeller, which provides the main engine thrust.

Propeller spins to provide engine thrust



Turbo-jet engines

Air is drawn in and compressed, then sent to a chamber where fuel burns. The gases produced are shot out of the back of the engine, which pushes the aircraft forwards, like a deflating balloon.

Gas shoots out

Air drawn in



Turbo-fan engines

A hybrid of turbo-jet and turbo-props, the turbo-fan engine sucks in air, which is combined with the backdraft from a fan, and also sends air around the engine, producing the same effect as a propeller.



Helicopters

Unlike most aircraft, which have fixed wings, a helicopter has a spinning rotor with two or more long, thin blades attached. When the blades spin round, they lift the helicopter straight up into the air. A helicopter can take off from almost anywhere and does not need to use airport runways. It can hover in one place, and fly backwards, forwards, and sideways. This makes it the most versatile of all aircraft; it is very useful for transport, surveillance, and rescue missions.

Flying controls

A helicopter pilot has three flying controls. The collective pitch lever changes the amount of lift produced by the main rotor. The cyclic pitch control makes the helicopter move forwards, backwards, or sideways. Rudder pedals make the helicopter turn left or right.



Lifting off

Before take-off, the main and tail rotors are speeded up. When the main rotor is turning fast enough, the pilot lifts the collective pitch lever to increase the tilt of the rotor blades. The tilt produces lift, and the helicopter takes off. The higher the lever is lifted, the faster the aircraft rises.



Moving away

The cyclic pitch control makes the helicopter move in the direction the control is pushed. It tilts the main rotor so that some of the rotor's lift pulls the helicopter along. Here, the pilot has pulled the control back to make the helicopter move backwards.



Rotor blades twist to control the helicopter's direction.

Landing skids in place of undercarriage

Turbo-shaft engine

Main rotor

Tail rotor stops fuselage spinning in opposite direction to main rotor.



Igor Ivan Sikorsky

Sikorsky (1889–1972) was born in Ukraine, where he became an aeronautical engineer. In 1919 he moved to the United States where he set up an aircraft factory. He designed the first practical helicopter, the VS-300, which first flew in 1939. The design had to be modified many times: at one point, the helicopter flew in every direction except forwards.